Haris Majeed Raja

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Summary

Aspiring AI/ML researcher with hands-on experience in computer vision and multimodal deep learning, including a final year project on deepfake detection using VLM-based reasoning pipelines. Currently working as an AI Engineer at MLBench, applying object detection and self-hosted language model fine-tuning to real-world problems. Seeking to pursue graduate research in computer vision or multimodal learning, with the goal of contributing to open problems in model robustness, interpretability, and efficient AI systems.

Research Experience

Multimodal Deepfake Detection using VLMs (Final Year Project) Aug 2025 – Jun 2026
Supervised by Dr. Waqas Sultani, Information Technology University

- Benchmarked state-of-the-art deepfake detectors (SBI, AltFreezing, Face Warping Detection) across FaceForensics++, Celeb-DF-v2, and HiDF, showing binary classifiers collapse to near-chance accuracy on high-quality deepfakes with no mechanism to explain predictions.
- Designed **ExDDV-Judge**, a three-stage detect, explain, judge pipeline replacing 11B/7B-parameter VLMs with lightweight alternatives (BLIP-2 T5-XL, LLaVA-Mini), cutting model size by over 80% while running on a single consumer GPU.
- Diagnosed and resolved a critical real-frame misclassification failure in prior work by designing a **Dual-DB routing system** with a fine-tuned CLIP encoder, raising real-frame accuracy from 0% to 83%.
- Connected explanation generation with automated verification by pairing BLIP-2 T5-XL with **DeepfakeJudge**, producing faithfulness-scored natural-language explanations for model predictions.

Experience

AI Engineer Jun 2026 – Present
MLBench Pvt Ltd

- Fine-tuned RF-DETR-Base for document field detection, reaching 0.9335 precision and 0.8496 recall, improving F1 from 0.8568 to 0.8896 over a client's existing three-model pipeline.
- Evaluated OCR backends (Tesseract, DocTR) for downstream extraction accuracy, diagnosing systematic errors caused by low OCR quality.
- Trained DINOv2-based classifiers for a rice grain quality and variety analysis system across multiple image resolutions.

Teaching Assistant – Software Engineering & Artificial Intelligence Aug 2025 – Jun 2026
Information Technology University

- Assisted in teaching core AI concepts including search algorithms, reasoning under uncertainty, and fundamentals of machine learning, alongside software engineering principles.
- Mentored student research and project groups on model selection, experiment design, and evaluation methodology.

Machine Learning Intern Jun 2025 – Aug 2025
Arch Technologies

- Developed and evaluated ML models including TF-IDF and Naive Bayes for text classification, and regression models for prediction tasks, using scikit-learn.

Projects

Legal AI Chatbot | *GitHub Link* July 2026

- Built a RAG-based legal assistant with project-scoped retrieval over LangChain and FAISS, with every generated answer grounded in and citing its source document.

Footfall Counter | *GitHub Link* June 2026

- Built a real-time people-counting system using YOLO/RT-DETR and ByteTrack for persistent multi-object tracking across occlusion, optimized to run entirely on CPU.

Technical Skills

AI/ML: PyTorch, HuggingFace Transformers, Computer Vision (YOLO, RF-DETR, DINOv2, Cellpose-SAM, CLIP, ByteTrack, OpenCV), LLM/VLM APIs (Groq, BLIP-2, Qwen-VL), LangChain, FAISS, RAG Pipelines, scikit-learn
Programming: Python, C++, Java, C, SQL
Tools: Git, Jupyter, Linux (Ubuntu)